FOR A NOETHERIAN DOMAIN, MOD_A^{FG} / T^{FG} IS CANONICALLY EQUIVALENT TO FINITE-DIMENSIONAL K-VECTOR SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The proof is terse by author style: the equivalence restriction is stated in one sentence and fullness is argued by the 'else torsion submodules ... might not be objects of T^{fg}' clause; content verbatim, no expansion.

Editorial note. The label proposition-quotient-by-finitely-generated-torsion-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Serre quotient Mod_A / T and its equivalence with K-vector spaces (same-chapter proposition-quotient-by-torsion-modules)
- Serre subcategories and quotient categories; full functors; Noetherian hypotheses making Mod_A^{fg} abelian (More on Algebra, section-abelian-categories-modules)

Source reference. Original section 0AZ5: sources/stacks-chow/tags.txt, source lines 13813–13813. Prerequisite pointer invoked at 6852; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B0K: sources/stacks-examples/pinned-original-source.tex, source lines 6818–6818. Prerequisite pointer invoked at 6847; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let A be a Noetherian integral domain. Let K denote its field of fractions. Let Mod_A^{fg} denote the category of finitely generated A-modules and $\mathcal{T}^{fg}$ its Serre subcategory of finitely generated torsion modules. Then $\mathit{Mod}_A^{fg}/\mathcal{T}^{fg}$ is canonically equivalent to the category of finite dimensional K-vector spaces.*

Proof. The equivalence given in Proposition 0B0K restricts along the embedding $\mathrm{Mod}_A^{fg}/\mathcal{T}^{fg} \to \mathrm{Mod}_A/\mathcal{T}$ to an equivalence $\mathrm{Mod}_A^{fg}/\mathcal{T}^{fg} \to \mathrm{Vect}_K^{fd}$. The Noetherian assumption guarantees that Mod_A^{fg} is an abelian category (see More on Algebra, Section 0AZ5) and that the canonical functor $\mathrm{Mod}_A^{fg}/\mathcal{T}^{fg} \to \mathrm{Mod}_A/\mathcal{T}$ is full (else torsion submodules of finitely generated modules might not be objects of $\mathcal{T}^{fg}$). □

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